irq: Firmware Starts but Never Reaches the Wait Loop

PHASE	TEST	STATUS
Phase 2	**irq**	**WAVEFORM**

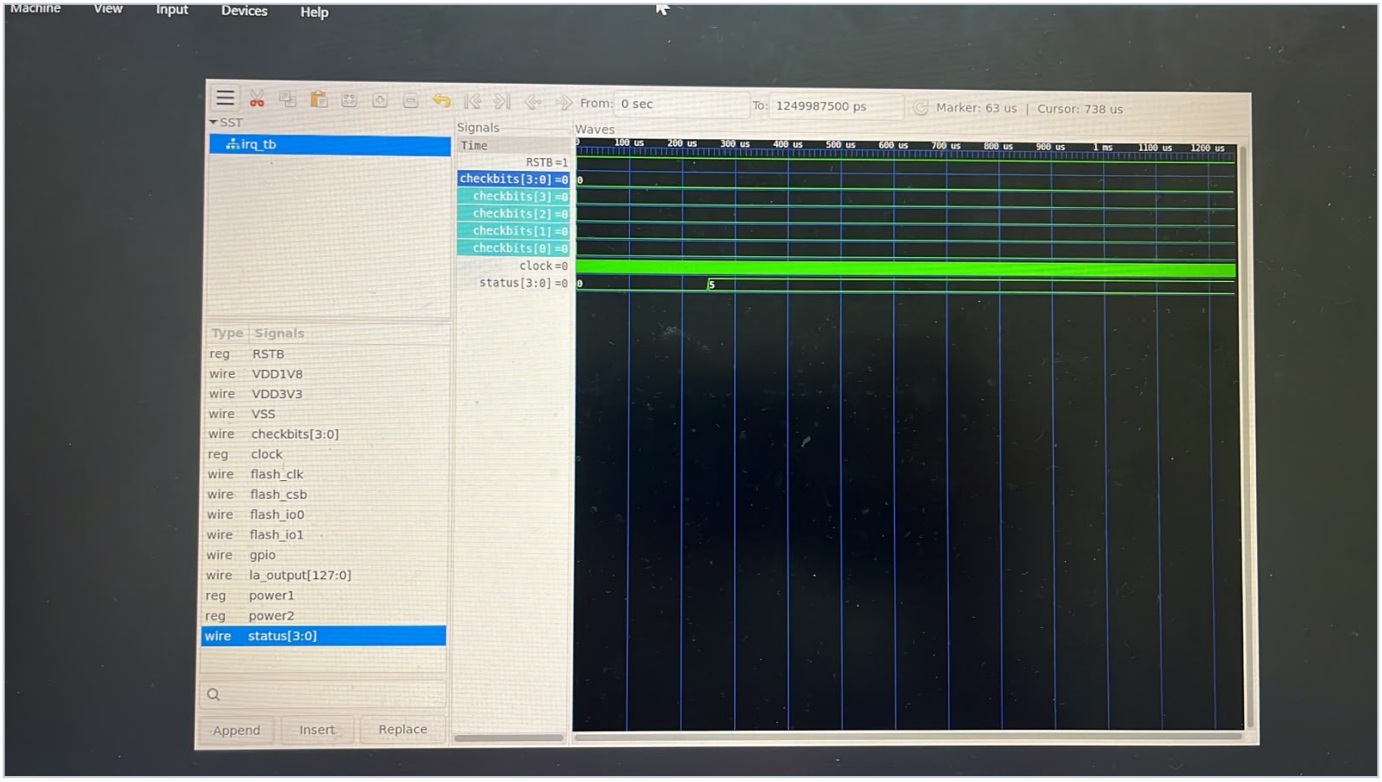


Figure 10. RTL-irq.vcd with RSTB, checkbits, clock and status

What this shows

- `status` becomes `5` at about 250 us: the firmware started (its start marker).
- `checkbits` stays `0` for the whole run. The firmware would set it to `1` once it entered the `while (flag == 0)` wait loop.
- Clock and reset are fine, so the problem is inside the firmware set-up, before the wait loop.

> **Conclusion.** Unlike timer, this is not a time-limit problem: the firmware stopped early.